

2 Mathematics (Specification A) content

Foundation Tier

1: Numbers and the number system	11
2: Equations, formulae and identities	15
3: Sequences, functions and graphs	17
4: Geometry and trigonometry	19
5: Vectors and transformation geometry	23
6: Statistics and probability	24

Higher Tier

1: Numbers and the number system	29
2: Equations, formulae and identities	31
3: Sequences, functions and graphs	33
4: Geometry and trigonometry	36
5: Vectors and transformation geometry	38
6: Statistics and probability	39

Foundation Tier

Externally assessed

Description

The Pearson Edexcel International GCSE in Mathematics (Specification A) requires students to demonstrate application and understanding of the following.

Number

- Use numerical skills in a purely mathematical way and in real-life situations.

Algebra

- Use letters as equivalent to numbers and as variables.
- Understand the distinction between expressions, equations and formulae.
- Use algebra to set up and solve problems.
- Demonstrate manipulative skills.
- Construct and use graphs.

Geometry

- Use properties of angles.
- Understand a range of transformations.
- Work within the metric system.
- Understand ideas of space and shape.
- Use ruler, compasses and protractor appropriately.

Statistics

- Understand basic ideas of statistical averages.
- Use a range of statistical techniques.
- Use basic ideas of probability.

Students should be able to demonstrate **problem-solving skills** by translating problems in mathematical or non-mathematical contexts into a process or a series of mathematical processes.

Students should be able to demonstrate **mathematical reasoning skills** by:

Assessment information

Each paper is assessed through a 2-hour examination set and marked by Pearson.

The total number of marks for each paper is 100.

Each paper will assess the full range of targeted grades at Foundation Tier (5–1).

Each paper will have approximately equal marks available for each of the targeted grades.

There will be approximately 40% of questions targeted at grades 5 and 4, across papers 1F and 1H to aid standardisation and comparability of award between tiers.

Diagrams will not necessarily be drawn to scale and measurements should not be taken from diagrams unless instructions to this effect are given.

Each student may be required to use mathematical instruments, e.g. pair of compasses, ruler, protractor.

A Foundation Tier formulae sheet (*Appendix 4*) will be included in the written examinations.

Tracing paper may be used in the examinations.

A calculator may be used in the examinations (please see *page 42* for further information).

Questions will be set in SI units (international system of units).